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Sample calculations, Chapter 5 (Savitsky and M-factor comparison): one worked example of each calculation type

1. Inputs (Table 5.1): LCG 640 mm, water at 15.8 °C

$b := 0.4\ \text{m}$ *Beam*

$W := 264.87\ \text{N}$ *Displacement weight*

$\beta := 18$ *Deadrise angle (degrees)*

$LCG := 0.64\ \text{m}$ *LCG from transom*

$VCG := 0.092\ \text{m}$ *VCG above keel*

Thrust inclination to keel (degrees)

$\varepsilon := 0$

Water density at 15.8 °C, ITTC (2011)

$\rho := 998.9783\ \frac{\text{kg}}{\text{m}^3}$

$g := 9.81\ \frac{\text{m}}{\text{s}^2}$ *Gravitational acceleration*

$\Delta C_f := 0.0004$ *ATTC roughness allowance*

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Thrust line to CG, negative above CG

$f := -0.0696\ \mathrm{m}$

Kinematic viscosity at 15.8 °C, ITTC (2011)

$\nu := 1.115 \cdot 10^{-6}\ \frac{\mathrm{m}^2}{\mathrm{s}} = 1.115 \cdot 10^{-6}\ \mathrm{s\ Gy}$

2. Speed and load coefficients at the measured point Fn∇ = 2.94

Measured Fn ∇ (step 10, mean of three sets)

$Fn := 2.940551$

√(g ∇^(1/3))

$V_{ref} := \sqrt{g \cdot \left(\frac{W}{\rho \cdot g}\right)^{\left(\frac{1}{3}\right)}} = 1.7158\ \frac{\mathrm{m}}{\mathrm{s}}$

$V := Fn \cdot V_{ref} = 5.0454\ \frac{\mathrm{m}}{\mathrm{s}}$

Speed

$C_v := \frac{V}{\sqrt{g \cdot b}} = 2.547$

Speed coefficient Cv = V/√(gb)

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Load coefficient CLβ = Δ/(½ρV²b²)

$$C_{Lb} := \frac{W}{0.5 \cdot \rho \cdot V^2 \cdot b^2} = 0.1302$$

3. Equilibrium trim from the code, and the lift equations checked at it

Equilibrium trim from Savitsky_Code_Test_m_Factor_All.m

$\tau := 4.878805576$

$\tau r := \frac{\tau \cdot \pi}{180} = 0.0852$ *Trim (rad)*

$\beta r := \frac{\beta \cdot \pi}{180} = 0.3142$ *Deadrise (rad)*

$\varepsilon r := \frac{\varepsilon \cdot \pi}{180} = 0$ *Thrust inclination (rad)*

Zero-deadrise lift coefficient, solved by the code

$C_{L0} := 0.1707007488$

Check: CLβ = CL0 − 0.0065 β CL0^0.6, residual ≈ 0

$$r_1 := C_{L0} - 0.0065 \cdot \beta \cdot C_{L0}^{0.6} - C_{Lb} = 1.0437 \cdot 10^{-12}$$

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Mean wetted length-to-beam ratio, solved by the code

$\lambda := 2.698961424$

Check: CL0 = τ^1.1 (0.012 λ^0.5 + 0.0055 λ^2.5 / Cv²), residual ≈ 0

$$r_2 := \tau^{1.1} \cdot \left(0.012 \cdot \lambda^{0.5} + \frac{0.0055 \cdot \lambda^{2.5}}{C_v^{2}} \right) - C_{L0} = -1.6687 \cdot 10^{-11}$$

4. Friction drag

Dynamic lift coefficient

$C_{Ld} := 0.012 \cdot \lambda^{0.5} \cdot \tau^{1.1} = 0.1127$

Mean bottom velocity

$$V_m := V \cdot \sqrt{1 - \frac{\left(C_{Ld} - 0.0065 \cdot \beta \cdot C_{Ld}^{0.6} \right)}{\lambda \cdot \cos(\tau r)}} = 4.9687 \ \frac{\mathrm{m}}{\mathrm{s}}$$

Reynolds number

$Re := \frac{V_m \cdot \lambda \cdot b}{\nu} = 4.8109 \cdot 10^{6}$

Schoenherr friction coefficient, solved by the code

$C_f := 0.003315556866$

Check: 0.242/√Cf = log10(Re Cf), residual ≈ 0

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$$r_3 := \frac{0.242}{\sqrt{C_f}} - \frac{\ln\left(Re \cdot C_f\right)}{\ln(10)} = -1.7018 \cdot 10^{-10}$$

Friction drag

$$D_f := \frac{\left(C_f + \Delta C_f\right) \cdot \rho \cdot V_m^{\,2} \cdot \lambda \cdot b^{\,2}}{2 \cdot \cos\left(\beta r\right)} = 20.8043 \text{ N}$$

5. Total drag

$$D := W \cdot \tan\left(\tau r\right) + \frac{D_f}{\cos\left(\tau r\right)} = 43.4886 \text{ N}$$

Drag D = Δ tan τ + Df / cos τ

$$DW := \frac{D}{W} = 0.1642$$

Predicted D/Δ (Savitsky)

6. Moment balance at the equilibrium trim

Centre-of-pressure coefficient

$$C_p := 0.75 - \frac{1}{\frac{5.21 \cdot C_v^{\,2}}{\lambda} + 2.39} = 0.6078$$

Distance from centre of pressure to CG

$$l_c := LCG - C_p \cdot \lambda \cdot b = -0.0161 \text{ m}$$

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$a := VCG - \frac{b}{4} \cdot \tan(\beta r) = 0.0595\ \text{m}$ *Friction-force arm*

Pitching-moment residual, ≈ 0 at equilibrium

$$M_{res} := W \cdot \left| \frac{(1 - \sin(\tau r) \cdot \sin(\tau r + \varepsilon r)) \cdot l_c}{\cos(\tau r)} - f \cdot \sin(\tau r) \right| + D_f \cdot (a - f) = 1.9135 \cdot 10^{-8}\ \text{J}$$

7. Blount and Fox (1976) M-factor

$r := \frac{LCG}{b} = 1.6$ *LCG / beam*

M-factor

$$M := 0.98 + 2 \cdot r^{1.45} \cdot \exp(-2 \cdot (Fn - 0.85)) - 3 \cdot r \cdot \exp(-3 \cdot (Fn - 0.85)) = 1.0314$$

$DW_M := M \cdot DW = 0.1693$ *Predicted D/Δ with the M-factor*

8. Prediction error (Equation 5.1)

$DW_{meas} := 0.171417$ *Measured D/Δ (mean of three sets)*

$\tau_{meas} := 5.381742$ *Measured trim (deg)*

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$$e_S := \frac{DW - DW_{meas}}{DW_{meas}} \cdot 100 = -4.2169$$ *Savitsky drag error (%)*

$$e_M := \frac{DW_M - DW_{meas}}{DW_{meas}} \cdot 100 = -1.214$$ *Savitsky + M-factor drag error (%)*

$$e_\tau := \frac{\tau - \tau_{meas}}{\tau_{meas}} \cdot 100 = -9.3452$$ *Trim error (%)*

9. Mean absolute error (Equation 5.2): Savitsky drag, LCG 640 mm, 8 compared points

$e_1 := -9_{748040286}$ *Point 1 drag error (%)*

$e_2 := -4_{334887397}$ *Point 2 drag error (%)*

$e_3 := -4_{383830343}$ *Point 3 drag error (%)*

$e_4 := -4_{119027373}$ *Point 4 drag error (%)*

$e_5 := -4_{216903278}$ *Point 5 drag error (%)*

$e_6 := -3_{891702261}$ *Point 6 drag error (%)*

$e_7 := -4_{149225416}$ *Point 7 drag error (%)*

$e_8 := -4_{342719486}$ *Point 8 drag error (%)*

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Mean absolute error (%)

$$MAE := \frac{|e_1| + |e_2| + |e_3| + |e_4| + |e_5| + |e_6| + |e_7| + |e_8|}{8} = 4.8983$$

10. Wetted lengths (Equations 5.3 and 5.4): LCG 560 mm, 4.10 m/s, water at 19.0 °C

Predicted trim (deg), Savitsky_Wetted_Area.m

$\tau_p := 6.652861816$

$\lambda_p := 2.447397914$ *Predicted λ*

Lk − Lc = b tan β / (π tan τ)

$$\Delta L := \frac{b \cdot \tan(\beta r)}{\pi \cdot \tan\left(\frac{\tau_p \cdot \pi}{180}\right)} = 0.3547 \text{ m}$$

$L_{kp} := \lambda_p \cdot b + \frac{\Delta L}{2} = 1.1563$ m *Predicted wetted keel length*

$L_{cp} := \lambda_p \cdot b - \frac{\Delta L}{2} = 0.8016$ m *Predicted wetted chine length*

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$L_k := 1.175\ \mathrm{m}$ *Measured wetted keel length*

$L_c := 0.845\ \mathrm{m}$ *Measured wetted chine length*

$\lambda_m := \frac{L_k + L_c}{2 \cdot b} = 2.525$ *Measured λ (Equation 5.3)*

$e_\lambda := \frac{\lambda_p}{\lambda_m} - 1 \cdot 100 = -3.0733$ *Error in λ, Equation 5.1 (%)*

Error in keel length, Equation 5.1 (%)

$e_{Lk} := \frac{L_{kp}}{L_k} - 1 \cdot 100 = -1.5914$

Error in chine length, Equation 5.1 (%)

$e_{Lc} := \frac{L_{cp}}{L_c} - 1 \cdot 100 = -5.1341$

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11. Savitsky drag at the measured trim (Section 5.6): LCG 640 mm, Fn∇ = 2.94

Mean wetted length-to-beam ratio at the measured trim τ.meas, solved by the code

$\lambda_{mt} := 2.4547478102$

Check: CL0 = τ^1.1 (0.012 λ^0.5 + 0.0055 λ^2.5 / Cv²), residual ≈ 0

$r_4 := \tau_{meas}^{1.1} \cdot \left| 0.012 \cdot \lambda_{mt}^{0.5} + \frac{0.0055 \cdot \lambda_{mt}^{2.5}}{C_v^2} \right| - C_{L0} = -1.6216 \cdot 10^{-12}$

$\tau_{mr} := \frac{\tau_{meas} \cdot \pi}{180} = 0.0939$ *Measured trim (rad)*

$C_{Ld2} := 0.012 \cdot \lambda_{mt}^{0.5} \cdot \tau_{meas}^{1.1} = 0.1197$ *Dynamic lift coefficient*

$V_{m2} := V \cdot \sqrt{1 - \frac{\left| C_{Ld2} - 0.0065 \cdot \beta \cdot C_{Ld2}^{0.6} \right|}{\left| \lambda_{mt} \cdot \cos\left(\tau_{mr} \right) \right|}} = 4.9548 \ \frac{m}{s}$ *Mean bottom velocity*

$Re_2 := \frac{V_{m2} \cdot \lambda_{mt} \cdot b}{\nu} = 4.3633 \cdot 10^{6}$ *Reynolds number*

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Schoenherr friction coefficient, solved by the code

$C_{f2} := 0.003371670355$

Check: 0.242/√Cf = log10(Re Cf), residual ≈ 0

$$r_5 := \frac{0.242}{\sqrt{C_{f2}}} - \frac{\ln\left(Re_2 \cdot C_{f2}\right)}{\ln(10)} = -2.5006 \cdot 10^{-10}$$

Friction drag

$$D_{f2} := \frac{\left(C_{f2} + \Delta C_f\right) \cdot \rho \cdot V_{m2}^{2} \cdot \lambda_{mt} \cdot b^{2}}{\left(2 \cdot \cos\left(\beta r\right)\right)} = 19.1001\ \text{N}$$

Drag at the measured trim

$$D_2 := W \cdot \tan\left(\tau_{mr}\right) + \frac{D_{f2}}{\cos\left(\tau_{mr}\right)} = 44.1371\ \text{N}$$

Error, Equation 5.1 (%); MAE 4.2% at 640 mm

$$e_{mt} := \frac{\left(\frac{D_2}{W} - DW_{meas}\right)}{DW_{meas}} \cdot 100 = -2.7886$$

12. Full-scale vessel (Section 5.5): scale factor, sea water and scaled Savitsky inputs

Length scale factor

$$SF := \frac{50}{1.6} = 31.25$$

Model water at 16.8 °C (LCG 560 mm runs), ITTC (2011)

$$\rho_m := \frac{998.8125\ \text{kg}}{3\ \text{m}}$$

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$\nu_m := \frac{1.0867 \cdot 10^{-6}\ m^2}{s}$ *Model water kinematic viscosity at 16.8 °C*

$\nabla_m := \frac{W}{(\rho_m \cdot g)} = 0.027\ m^3$ *Model displaced volume*

$\nabla_s := \nabla_m \cdot SF^3 = 824.9542\ m^3$ *Vessel displaced volume (SF³)*

$\rho_s := \frac{1026\ kg}{m^3}$ *Sea water density*

$\mu_s := 0.00122\ Pa\ s$ *Sea water dynamic viscosity*

$\nu_s := \frac{\mu_s}{\rho_s} = 1.1891 \cdot 10^{-6}\ s\ Gy$ *Sea water kinematic viscosity*

$\Delta_s := \rho_s \cdot g \cdot \nabla_s = 8.3032 \cdot 10^6\ N$ *Vessel displacement (8303 kN)*

$b_s := b \cdot SF = 12.5\ m$ *Beam*

$LCG_s := 0.56\ m \cdot SF = 17.5\ m$ *LCG from transom*

$VCG_s := VCG \cdot SF = 2.875\ m$ *VCG above keel*

$f_s := f \cdot SF = -2.175\ m$ *Thrust line to CG*

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$L_{ps} := 1.6\ \text{m} \cdot SF = 50\ \text{m}$ *Planing length*

$V_{refs} := \sqrt{g \cdot \nabla_s^{\left(\frac{1}{3}\right)}} = 9.5919\ \frac{\text{m}}{\text{s}}$ *√(g ∇^(1/3)) of the vessel*

13. ITTC 1957 extrapolation (Equations 5.5–5.12): Run 8, LCG 560 mm, the thrust peak

$V_{m8} := \frac{3.99944297075272\ \text{m}}{\text{s}}$ *Measured model speed, Run 8*

$R_{Tm} := 47.7120614242554\ \text{N}$ *Measured model resistance, Run 8*

$V_{ref5} := \sqrt{g \cdot \nabla_m^{\left(\frac{1}{3}\right)}} = 1.7159\ \frac{\text{m}}{\text{s}}$ *√(g ∇^(1/3)) of the model at 16.8 °C*

$Fr := \frac{V_{m8}}{V_{ref5}} = 2.3309$ *Fr = Fn ∇*

Regression coefficients (Equations 5.11 and 5.12, wetted-area runs)

$a_0 := 0.9050494603602045$

$a_1 := -0.2764890609430355$

$a_2 := 0.03285516720684278$

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$c_0 := 1.951194206411187$

$c_1 := -0.3079738200561940 6$

$S_m := \left(a_0 + a_1 \cdot Fr + a_2 \cdot Fr^2\right) m^2 = 0.4391\ m^2$ *Model wetted surface, Equation 5.11*

$L_{km} := \left(c_0 + c_1 \cdot Fr\right) m = 1.2333\ m$ *Model wetted keel length, Equation 5.12*

$Rn_m := \frac{V_{m8} \cdot L_{km}}{\nu_m} = 4.5392 \cdot 10^6$ *Model Reynolds number, Equation 5.7*

$c_{Tm} := \frac{R_{Tm}}{\left(0.5 \cdot \rho_m \cdot V_{m8}^2 \cdot S_m\right)} = 0.0136$ *Equation 5.5*

$c_{Fm} := \frac{0.075}{\left(\frac{\ln\left(Rn_m\right)}{\ln(10)} - 2\right)^2} = 0.0035$ *ITTC 1957 line, Equation 5.6*

$c_R := c_{Tm} - c_{Fm} = 0.0101$ *Residual coefficient, Equation 5.8*

$V_s := V_{m8} \cdot \sqrt{SF} = 22.3576\ \frac{m}{s}$ *Vessel speed (Froude scaling)*

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$$V_{kn} := \frac{V_s}{\left(\frac{0.514444\ \text{m}}{\text{s}}\right)} = 43.4597$$

Vessel speed (knots)

$$L_{ks} := L_{km} \cdot SF = 38.5421\ \text{m}$$

Vessel wetted keel length

$$S_s := S_m \cdot SF^2 = 428.7987\ \text{m}^2$$

Vessel wetted surface

$$Rn_s := \frac{V_s \cdot L_{ks}}{\nu_s} = 7.2468 \cdot 10^8$$

Vessel Reynolds number

$$Rn_{ratio} := \frac{Rn_s}{Rn_m} = 159.6512$$

≈ 160 times higher

$$c_{Fs} := \frac{0.075}{\left(\frac{\ln\left(Rn_s\right)}{\ln\left(10\right)} - 2\right)^2} = 0.0016$$

Equation 5.6, vessel

$$c_A := 0.00035 - 2 \cdot 50 \cdot 10^{-6} = 0.0002$$

Correlation allowance, L = 50 m

$$c_{Ts} := c_R + c_{Fs} + c_A = 0.012$$

Equation 5.9

$$R_{Ts} := c_{Ts} \cdot 0.5 \cdot \rho_s \cdot V_s^2 \cdot S_s = 1.3182 \cdot 10^6\ \text{N}$$

Equation 5.10: required thrust (1318 kN)

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Regression limit: Fr at which the fitted keel length equals the 1.6 m planing length

$$Fr_{Lp} := \frac{\left(c_0 - 1.6\right)}{\left(-c_1\right)} = 1.1403$$

Below this, Lk > Lp (Runs 1 to 4)

14. Full-scale Savitsky prediction at the same Fr (Savitsky_FullScale_vs_ITTC.m)

$$V_S := Fr \cdot V_{refs} = 22.3576 \ \frac{m}{s}$$

Equals V.s: same Froude number

$$C_{vs} := \frac{V_S}{\sqrt{g \cdot b_s}} = 2.019$$

Speed coefficient

$$C_{Lbs} := \frac{\Delta_s}{\left(0.5 \cdot \rho_s \cdot V_S^{\ 2} \cdot b_s^{\ 2}\right)} = 0.2072$$

Load coefficient

Equilibrium trim, CL0, λ and Cf, solved by the code

$$\tau_s := 6.7662767222$$

Trim (deg)

$$C_{L0s} := 0.2592875072$$

$$\lambda_s := 2.4614201492$$

$$C_{fs} := 0.001644408182$$

$$\tau_{sr} := \frac{\tau_s \cdot \pi}{180} = 0.1181$$

Trim (rad)

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$r_6 := C_{L0s} - 0.0065 \cdot \beta \cdot C_{L0s}^{0.6} - C_{Lbs} = -8.3164 \cdot 10^{-12}$ *Check, Equation 6: ≈ 0*

$r_7 := \tau_s^{1.1} \cdot \left(0.012 \cdot \lambda_s^{0.5} + \frac{0.0055 \cdot \lambda_s^{2.5}}{C_{vs}^2} \right) - C_{L0s} = 5.582 \cdot 10^{-12}$ *Check, Equation 7: ≈ 0*

$C_{Lds} := 0.012 \cdot \lambda_s^{0.5} \cdot \tau_s^{1.1} = 0.1542$ *Dynamic lift coefficient*

$V_{ms} := V_S \cdot \sqrt{1 - \frac{\left(C_{Lds} - 0.0065 \cdot \beta \cdot C_{Lds}^{0.6} \right)}{\left(\lambda_s \cdot \cos\left(\tau_{sr} \right) \right)}} = 21.8201 \ \frac{m}{s}$ *Mean bottom velocity*

$Re_s := \frac{V_{ms} \cdot \lambda_s \cdot b_s}{\nu_s} = 5.646 \cdot 10^8$ *Reynolds number*

$r_8 := \frac{0.242}{\sqrt{C_{fs}}} - \frac{\ln\left(Re_s \cdot C_{fs} \right)}{\ln(10)} = -1.4556 \cdot 10^{-10}$ *Check, Schoenherr: ≈ 0*

$D_{fs} := \frac{\left(C_{fs} + \Delta C_f \right) \cdot \rho_s \cdot V_{ms}^2 \cdot \lambda_s \cdot b_s^2}{\left(2 \cdot \cos\left(\beta r \right) \right)} = 2.0193 \cdot 10^5 \ N$ *Friction drag*

$D_s := \Delta_s \cdot \tan\left(\tau_{sr} \right) + \frac{D_{fs}}{\cos\left(\tau_{sr} \right)} = 1.1885 \cdot 10^6 \ N$ *Savitsky resistance (1188 kN)*

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$$C_{ps} := 0.75 - \frac{1}{\left(\frac{5.21 \cdot C_{vs}^{2}}{\lambda_s^{2}} + 2.39\right)} = 0.5804$$

Centre-of-pressure coefficient

$$l_{cs} := LCG_s - C_{ps} \cdot \lambda_s \cdot b_s = -0.3569\ \text{m}$$

Centre of pressure to CG

Friction-force arm

$$a_s := VCG_s - \frac{b_s}{4} \cdot \tan(\beta r) = 1.8596\ \text{m}$$

Moment residual ≈ 0 (relative to Δ.s·b.s)

$$M_{ress} := \Delta_s \cdot \left| \frac{\left(1 - \sin(\tau_{sr}) \cdot \sin(\tau_{sr} + \varepsilon r)\right) \cdot l_{cs}}{\cos(\tau_{sr})} - f_s \cdot \sin(\tau_{sr}) \right| + D_{fs} \cdot (a_s - f_s) = 0.0008\ \text{J}$$

LCG / beam = 1.4, as at model scale

$$r_s := \frac{LCG_s}{b_s} = 1.4$$

M-factor, Equation 3.19

$$M_s := 0.98 + 2 \cdot r_s^{1.45} \cdot \exp(-2 \cdot (Fr - 0.85)) - 3 \cdot r_s \cdot \exp(-3 \cdot (Fr - 0.85)) = 1.0991$$

Savitsky + M-factor resistance (1306 kN)

$$D_{sM} := M_s \cdot D_s = 1.3063 \cdot 10^{6}\ \text{N}$$

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15. Difference from the ITTC 1957 thrust (Equation 5.1, ITTC 1957 as reference)

$$e_{SI} := \frac{\left(D_S - R_{TS}\right)}{R_{TS}} \cdot 100 = -9.8373$$ *Savitsky (%)*

$$e_{MI} := \frac{\left(D_{SM} - R_{TS}\right)}{R_{TS}} \cdot 100 = -0.9018$$ *Savitsky + M-factor (%)*

Mean absolute difference, bare Savitsky, 10 points with Fr ≥ 1.0 (Table 5.4)

$e_{s1} := -11.4373933576$

$e_{s2} := -13.5768807461$

$e_{s3} := -8.3262106330$

$e_{s4} := -9.8149265951$

$e_{s5} := -9.8373637015$

$e_{s6} := -9.0237461696$

$e_{s7} := -9.7795336818$

$e_{s8} := -10.2913868817$

$e_{s9} := -11.5078128492$

$e_{s10} := -11.4857642223$

Mean absolute difference (%)

$$MAD_S := \frac{\left(\left|e_{s1}\right| + \left|e_{s2}\right| + \left|e_{s3}\right| + \left|e_{s4}\right| + \left|e_{s5}\right| + \left|e_{s6}\right| + \left|e_{s7}\right| + \left|e_{s8}\right| + \left|e_{s9}\right| + \left|e_{s10}\right|\right)}{10} = 10.5081$$

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16. Thrust analysis: thrust-to-weight ratio, effective power and friction share (Section 5.5)

$TW := \frac{R_{Ts}}{\Delta_s} = 0.1588$ — *Thrust-to-weight ratio at the peak (0.159)*

$P_E := R_{Ts} \cdot V_s = 2.9471 \cdot 10^{7}\ \mathrm{W}$ — *Effective power at the peak (29.5 MW)*

$R_{T11} := 1256624.04178926\ \mathrm{N}$ — *ITTC 1957 thrust, Run 11 (Fr = 3.23)*

$\delta_T := \frac{\left(R_{T11} - R_{Ts}\right)}{R_{Ts}} \cdot 100 = -4.6682$ — *Fall after the peak (%)*

$R_{T13} := 1255523.46493913\ \mathrm{N}$ — *ITTC 1957 thrust, Run 13 (Fr = 3.83)*

$V_{s13} := \frac{36.7223080709757\ \mathrm{m}}{\mathrm{s}}$ — *Vessel speed, Run 13 (71.4 knots)*

$P_{E13} := R_{T13} \cdot V_{s13} = 4.6106 \cdot 10^{7}\ \mathrm{W}$ — *Effective power at top speed (46.1 MW)*

$k_I := \frac{\left(c_{Fs} + c_A\right)}{c_{Ts}} \cdot 100 = 15.3791$ — *Friction share of the ITTC 1957 resistance at the peak (%)*

$k_S := \frac{\frac{D_{fs}}{\cos\left(\tau_{sr}\right)}}{D_s} \cdot 100 = 17.1095$ — *Friction share of the Savitsky resistance at the peak (%)*

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17. Difference in D/Δ at model and full scale at Fr = 2.33 (Section 5.6)

$DW_{m8} := \frac{R_{Tm}}{W} = 0.1801$ *Measured model D/Δ, LCG 560 mm*

$DW_{S8} := 0.16457302147$ *Savitsky model D/Δ at this Fr (Ch5 comparison)*

$g_m := DW_{m8} - DW_{S8} = 0.0156$ *Model-scale difference in D/Δ*

$g_s := \frac{R_{Ts}}{\Delta_s} - \frac{D_s}{\Delta_s} = 0.0156$ *Full-scale difference in D/Δ (0.0156 at both scales)*

18. Speed coefficient at the M-factor limit Fn∇ = 1.0 (Section 5.2)

$C_{v1} := \frac{1.0 \cdot V_{ref}}{\sqrt{g \cdot b}} = 0.8662$ *Cv ≥ 0.87 for Fn ∇ ≥ 1.0*

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